

Lost in Legalese: NLP for Privacy Risk Detection

CS224N Custom Project

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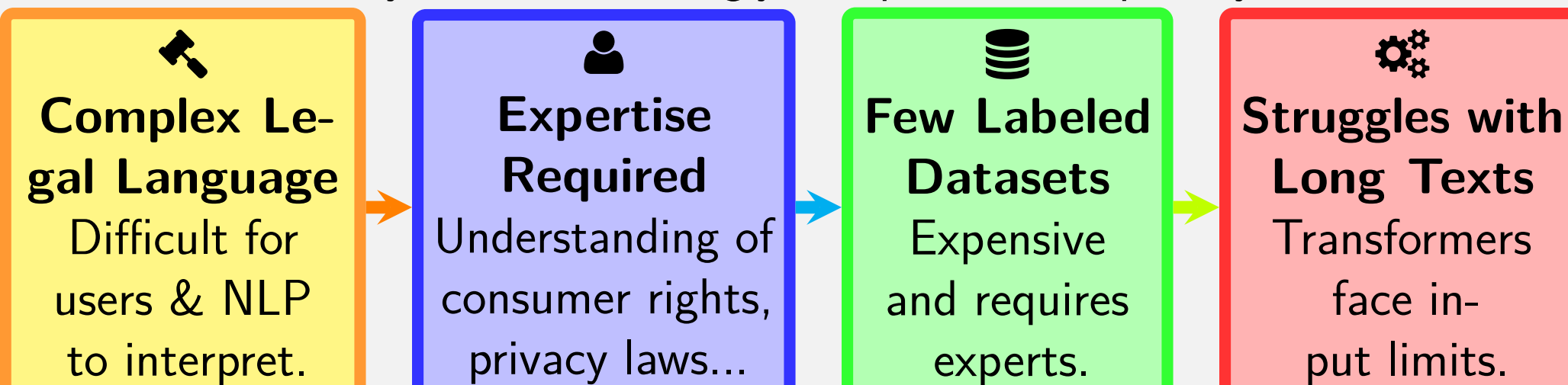
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Problem Statement

Legal T&C agreements are long, complex & obscure critical information. Many users unknowingly accept unfair or privacy-invasive clauses.



Background (Costly Mistake: 2019 Disney+ Case)



Figure 1. A hidden clause in a Disney+ agreement waived legal claims.

Existing NLP research focuses on court cases & contracts, not T&Cs. Most work does not assess fairness or privacy risks in consumer agreements.

Past Work Limitations:

- LEGAL-BERT is powerful for legal tasks but not optimized for fairness classification [1].
- Prior research focused on sentiment analysis, not privacy assessment.

Background:

A woman's family lost a lawsuit due to a **hidden T&C clause** in the 2019 Disney+ trial agreement.



Outcome: Disney **avoided liability** using a buried waiver clause that the user unknowingly accepted.

Approach

We fine-tune LEGAL-BERT using **Low-Rank Adaptation (LoRA)** [3] for fairness classification. Our model performs two key tasks:

Task 1: Individual Clause Classification

Classify each clause in T&C documents into: **"Very Bad"**, **"Bad"**, **"Neutral"**, **"Good"**

Task 2: Overall Document Scoring

Provide an **overall privacy risk score** (from **A** to **E**) for entire T&C documents.

Baseline: Zero-shot & Chain of Thought with LLaMA 2 (7B) [4].

Example Input & Model Prediction

"We may share your personal data with third-party partners without explicit consent."

Prediction: **Very Bad**
Ground Truth: **Very Bad**

Methods

LoRA: Parameter-Efficient Fine-Tuning

Why LoRA?

- Standard fine-tuning updates **ALL** model weights (345M parameters in BERT).
- LoRA updates only low-rank matrices ($r \ll d$), $\downarrow$ computation by **50%+** & $\uparrow$ generalization across unseen legal clauses.

Optimization Objective for LoRA Fine-Tuning:

$$\max_{\Theta, |\Theta| \ll |\Phi_0|} \sum_{(x,y) \in Z} \sum_{t=1}^{|y|} \log \left(p_{\Phi_0 + \Delta\Phi(\Theta)}(y_t | x, y_{<t}) \right)$$

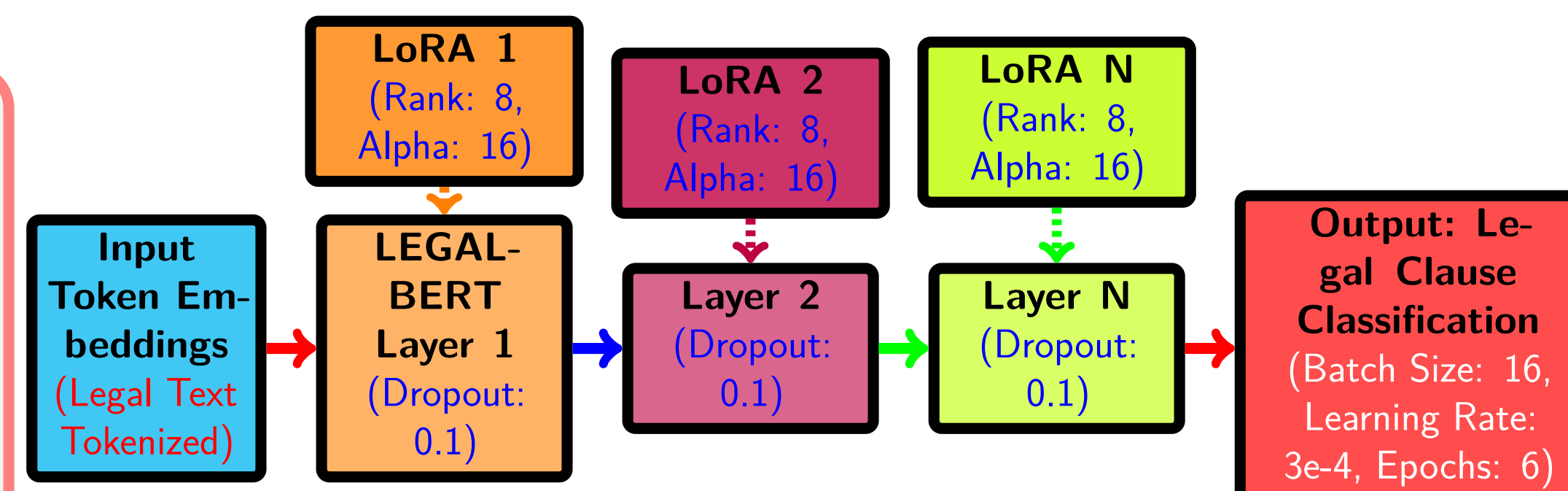


Figure 2. LEGAL-BERT + LoRA Model Architecture

Experiments & Results

Dataset	LLaMA 7B Zero-Shot	LLaMA 7B Chain of Thought
Accuracy on 5 post-08/2023 T&Cs	28.30%	30.29%
Accuracy on Facebook T&Cs	21.74%	23.91%
Accuracy on clear language	100.00%	100.00%

LLaMA 7B Zero-Shot & Chain of Thought Accuracy on different datasets

Legal NLP Challenges: While LLaMA 2 handles clear clauses with ease, its low accuracy on real-world legal text (<30%) highlights fundamental limitations.

Key Takeaways

Dataset: 450 T&C documents, 9,292 annotated clauses (human-labeled)

Challenges:

- "Very Bad" clauses under-predicted $\rightarrow$ Model hesitates on extreme cases.
- Legal cross-references $\rightarrow$ Issues with clauses citing external laws.
- Document scoring errors $\rightarrow$ Parsing mistakes impact fairness ratings.

Performance Highlights:

- ☒ Fine-tuned LegalBERT $\rightarrow$ Best for privacy risk classification
- ☒ LoRA $\rightarrow$ Boosts efficiency, reduces overfitting
- ☒ LLaMA models $\rightarrow$ Struggle with legal text comprehension
- ☒ BERT-based models $\rightarrow$ Offer better understanding of legal context

Analysis

Metrics	Zero-shot	Chain of Thought	LegalBERT (FT)	LegalBERT (LoRA)
Train Accuracy	-	-	0.97	0.86
Train Precision	-	-	0.97	0.86
Train Recall	-	-	0.97	0.86
Train F1-score	-	-	0.97	0.85
Test Accuracy	0.28	0.30	0.84	0.77
Test Precision	0.32	0.37	0.84	0.77
Test Recall	0.28	0.30	0.75	0.77
Test F1-score	0.23	0.24	0.84	0.77

Table 1. Comparing privacy risk prediction performance across architectures

Train-Test Contamination in Legal NLP

Why is this a problem?

- Legal documents often contain **paraphrased clauses**, making an exact train-test split insufficient.
- Evaluating on paraphrased clauses may **artificially inflate model performance**.

De-duplication Strategy: N-gram Similarity

- We measure clause similarity using **n-gram similarity** ($n = 3$) to balance:
 - Capturing meaningful relationships in legal language.
 - Avoiding excessive sensitivity to minor wording variations.
- Test clauses with **n-gram similarity** > 0.5 to a training clause are **removed**.

Critical Challenges

- Missed High-Risk Clauses – **48% recall**, failing to flag key risks.
- Legal Complexity – Struggles with law references.
- Unreliable Fairness Scores – Parsing errors (**27.14%** accuracy).

What's Next?

- Boost fairness recall – Adjust thresholds, confidence calibration.
- Enhance document scoring – Explore attention-based + graph-based aggregation.
- Legal Retrieval (RAG) [2] – Provide external legal context.
- Expand dataset – Finance + Healthcare contracts for generalization.

References

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